Clustering and annotating 2,700 PBMCs: a single-cell RNA-seq workflow, with a check on how stable the clusters are

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1 Summary

I analysed the public 10x Genomics PBMC 3k dataset (2,700 droplets, 32,738 genes) in Python with Scanpy (Wolf et al., 2018), following the pre-processing recommendations of Luecken & Theis (2019) and clustering with the Leiden algorithm (Traag et al., 2019). After quality control 2,434 cells remain. Leiden clustering at resolution 0.5 gives 8 clusters, which I annotated as 7 cell types from canonical marker genes: CD4 T, CD8 T, NK, B, CD14 and CD16 (FCGR3A) monocytes, and dendritic cells. My clusters agree moderately with the Scanpy tutorial's own Louvain clusters for the same cells (adjusted Rand index 0.613; the tutorial labels are a reference, not ground truth), and the per-type overlap is high for most types. When I re-cluster random 80% subsamples, the partition is moderately stable (mean ARI 0.749 against the full-data clustering, SD 0.077). Two of my first annotation attempts were wrong and are reported under "Two annotation attempts that failed". An independent classifier trained on external data (CellTypist) names all 8 clusters consistently with my rules, and is least confident on the CD8 T cluster – the one an earlier attempt of mine mislabelled.

2 Question

Can a standard workflow recover the main immune cell types in a small, well-known PBMC dataset, and how much do the clusters depend on choices the analyst makes: filtering rules, number of principal components, clustering resolution and random subsampling?

3 Data and methods

Data. `scanpy.datasets.pbmc3k()`, the public 10x Genomics "3k PBMCs from a healthy donor" count matrix, obtained through Scanpy. No data were generated for this report.

Quality control. I used adaptive (median-absolute-deviation) thresholds instead of fixed cut-offs: a cell is removed if its log total counts or log genes detected is more than 5 MADs from the median, or its mitochondrial fraction is more than 3 MADs above the median or above 20%. These choices follow Luecken & Theis (2019). For comparison I also computed how many cells the tutorial's fixed thresholds (200 to 2,500 genes, under 5% mitochondrial) would remove.

Processing. Library-size normalisation to 10,000 counts per cell, log(1 + x), 2,000 highly variable genes (Seurat flavour), scaling, PCA (50 components; the first 15 used), a 15-nearest-neighbour graph, UMAP for display, and Leiden clustering at resolutions 0.3, 0.5, 0.8 and 1.2.

Annotation. An ordered rule table on the fraction of a cluster's cells that express canonical markers (for example `MS4A1` for B cells, `FCER1A` for dendritic cells, `GNLY` for NK cells). The rules were written after inspecting marker fractions, so the annotation is not blind to the data. Per-cluster top genes (Wilcoxon rank-sum test) are reported as supporting evidence.

Stability. I drew 20 random subsamples of 80% of the cells, re-ran the neighbour graph and Leiden clustering, and compared each result with the full-data clustering on the shared cells using the adjusted Rand index (ARI).

Software. Scanpy 1.12.4, AnnData 0.13, python-igraph, scikit-learn, in a conda environment; random seed 0 throughout.

4 Results

4.1 Quality control

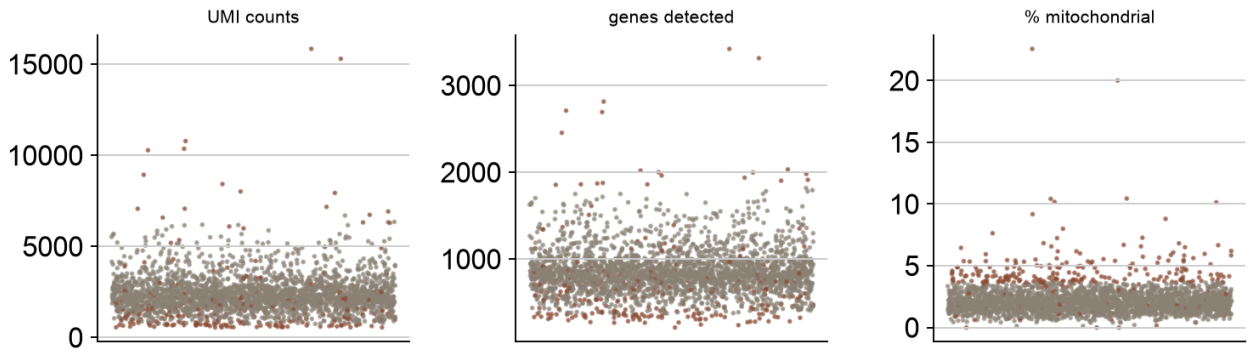


Figure 1: Quality-control metrics per droplet. Rust points are removed by the adaptive thresholds.

The adaptive rules remove 266 of 2,700 droplets, against 62 for the tutorial's fixed thresholds; all 62 cells flagged by the fixed rules are also flagged by the adaptive ones. The adaptive rules are therefore considerably stricter on this dataset. I did not test whether the extra 204 removed cells are low quality or simply have unusually low counts of a genuine small cell population, which is a real risk when filtering on count depth.

4.2 Clusters and annotation

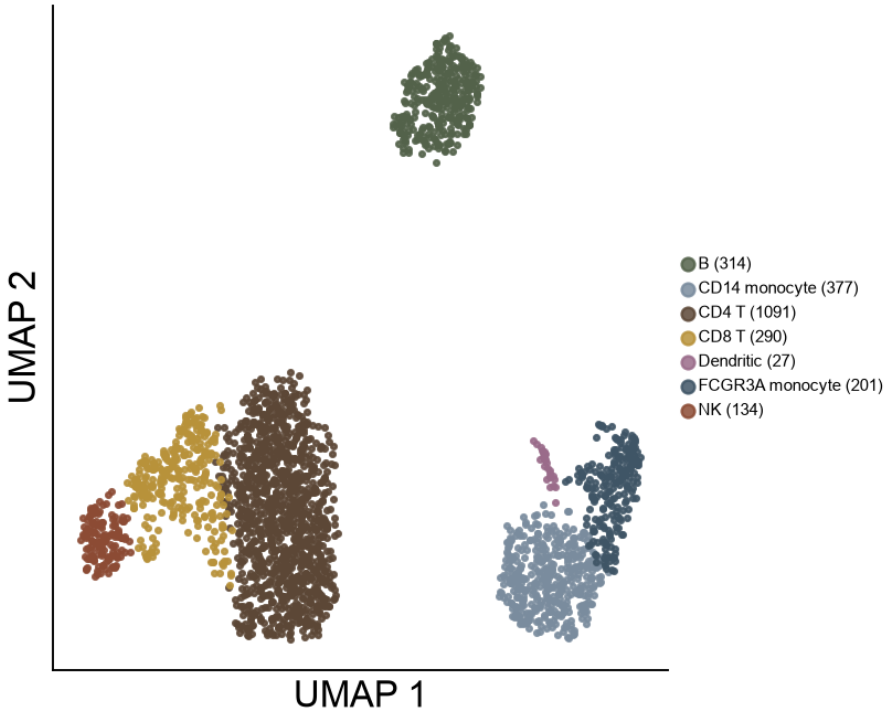


Figure 2: UMAP of the `r n(r$n_cells)` retained cells coloured by annotated cell type (Leiden, resolution 0.5).

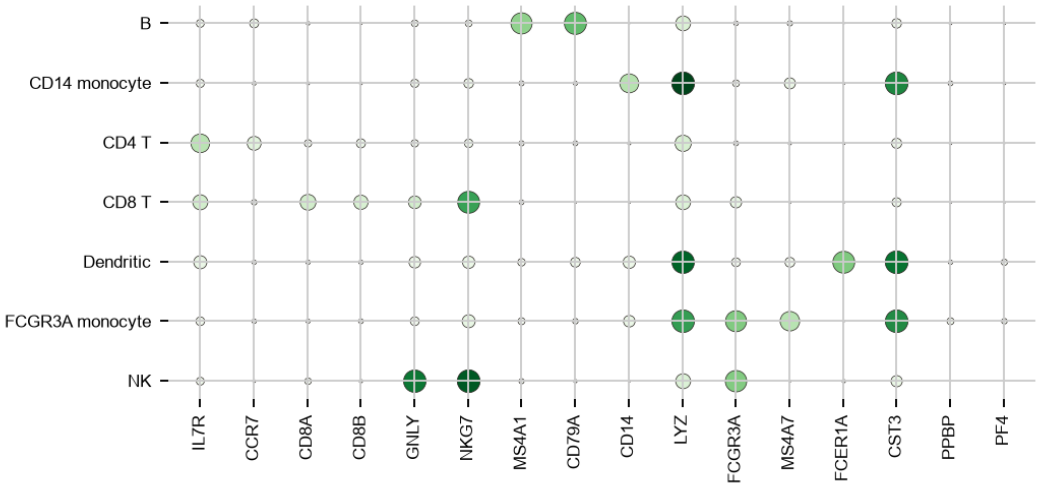


Figure 3: Canonical marker genes by annotated cell type. Dot size: fraction of cells expressing; shade: mean normalised expression.

Table 1: Cells per annotated type.

	Cells
CD4 T	1091
CD14 monocyte	377
B	314
CD8 T	290
FCGR3A monocyte	201
NK	134
Dendritic	27

Table 2: Top-ranked genes per Leiden cluster (Wilcoxon) and the label assigned.

Cluster	Label	Top genes
0	CD4 T	LTB, IL32, IL7R, TPT1, JUNB, RPS27A
1	CD4 T	RPL21, RPS3A, RPL13, MALAT1, RPS27A, RPS4X
2	CD8 T	CCL5, NKG7, CST7, B2M, GZMA, CTSW
3	FCGR3A monocyte	LST1, COTL1, AIF1, FCER1G, FTH1, SAT1
4	NK	NKG7, GZMB, PRF1, GNLY, CTSW, GZMA
5	B	CD74, CD79A, HLA-DRA, CD79B, HLA-DPB1, MS4A1
6	CD14 monocyte	S100A9, LYZ, S100A8, TYROBP, FCN1, LGALS2
7	Dendritic	FCER1A, HLA-DPB1, HLA-DPA1, HLA-DRA, CST3, HLA-DRB1

The two CD4 T clusters differ mainly in ribosomal and `CCR7` expression, which suggests a naive and a memory-like population, but I did not test that and keep them under one label. The label for cluster 3 (CD16 monocytes) rests on `FCGR3A` in 86% and `MS4A7` in 74% of its cells together with top genes `LST1`, `AIF1` and `FCER1G`.

4.3 Agreement with the tutorial's labels

Table 3: Overlap with the Scanpy tutorial's labels for the same cells (pairs with at least 5 cells).

Mine	Tutorial	Cells
B	B cells	312
CD14 monocyte	CD14+ Monocytes	374
CD4 T	CD4 T cells	1041
CD4 T	CD8 T cells	49
CD8 T	CD4 T cells	39
CD8 T	CD8 T cells	235
CD8 T	NK cells	16
Dendritic	Dendritic cells	27
FCGR3A monocyte	CD14+ Monocytes	54
FCGR3A monocyte	FCGR3A+ Monocytes	139
NK	NK cells	132

Agreement is high for B cells, dendritic cells, CD14 monocytes and NK cells (Table 3). The main differences are at boundaries: 54 cells I call CD16 monocytes are CD14 monocytes in the tutorial, and 39 of my CD8 T cells are CD4 T cells there. The tutorial also separates megakaryocytes (a few cells); at resolution 0.5 my clustering does not, so they sit inside the CD16 monocyte cluster. That is a limitation of the resolution I chose.

4.4 How stable are the clusters?

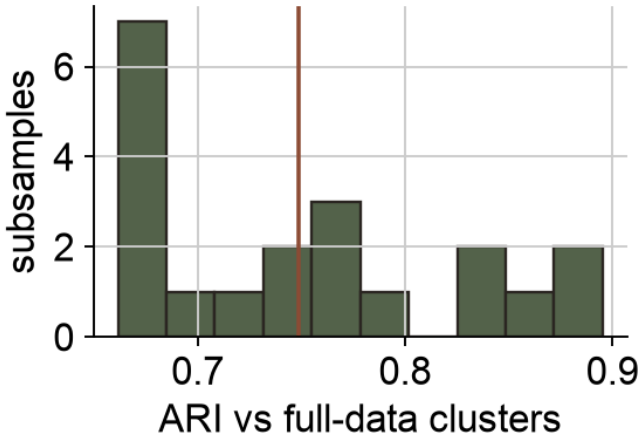


Figure 4: Stability of the resolution-0.5 clustering under 80% subsampling (20 repeats).

The number of clusters changes with resolution (0.3: 5; 0.5: 8; 0.8: 8; 1.2: 10 clusters), while resolutions 0.5 and 0.8 give very similar partitions (ARI 0.821). Subsampling gives a mean ARI of 0.749: the broad structure is reproducible, but individual cluster boundaries, particularly within the T-cell compartment, are not fixed. Cluster labels from a single run should not be treated as exact.

4.5 Two annotation attempts that failed

1. **Z-scoring cluster means** of marker genes across clusters labelled the CD16 monocyte cluster (201 cells) as megakaryocytes, on a margin of 0.2 and with only 9% of its cells expressing `PPBP`. Megakaryocytes are rare in PBMCs, so the label was implausible; the z-scoring was amplifying differences in barely expressed genes.
2. **An argmax over marker-panel scores** labelled the CD8 T cluster as NK cells, because `NKG7` is high in both, while `CD8A` and `CD8B` are poorly detected in this dataset.

I replaced both with the rule table above and added an assertion that fails if any cluster matches zero or more than one rule; that assertion caught a CD4/CD8 overlap on the next run. Because the rules were written after seeing the data, the annotation is a defensible expert labelling rather than an independent test; the next section supplies one.

4.6 Does an independent classifier agree?

The rule table was written after looking at marker fractions, so on its own it is an expert labelling rather than a test. CellTypist (Domínguez Conde et al., 2022) is trained on external annotated immune data and never sees this dataset, so it can disagree. I ran its Immune_All_Low.pkl model on the same normalised counts and compared, imposing no mapping between its label set and mine.

Table 4: Each cluster's most common CellTypist label against the label my rules assigned. No mapping between the two label sets is imposed.

Cluster	Cells	My rule label	CellTypist, most common	Share	Next most common
0	673	CD4 T	Tcm/Naive helper T cells	0.49	Tem/Effector helper T cells
1	418	CD4 T	Tcm/Naive helper T cells	0.63	Tcm/Naive cytotoxic T cells
2	290	CD8 T	Tem/Trm cytotoxic T cells	0.31	MAIT cells
3	201	FCGR3A monocyte	Non-classical monocytes	0.89	Classical monocytes

Cluster	Cells	My rule label	CellTypist, most common	Share	Next most common
4	134	NK	CD16+ NK cells	0.96	Tem/Temra cytotoxic T cells
5	314	B	B cells	0.95	Naive B cells
6	377	CD14 monocyte	Classical monocytes	0.87	Non-classical monocytes
7	27	Dendritic	DC	0.48	DC2

All 8 clusters are named consistently. Classical monocytes correspond to my CD14 cluster and non-classical to my CD16 cluster; cytotoxic T labels fall on the cluster I call CD8 T, helper T labels on the two I call CD4 T, and B, NK and dendritic agree directly (Table 4). That is the independent check the annotation previously lacked.

Its confidence varies a lot, and is lowest on the cluster this report already treated as the hardest. The dominant label covers 31% to 96% of a cluster's cells. The two lowest are cluster 2 (CD8 T, 31%) and the dendritic cluster, which has only 27 cells. The CD8 T cluster is the one an earlier attempt mislabelled as NK, and `CD8A` and `CD8B` are poorly detected here; CellTypist's hesitation on the same cluster is independent support for treating that boundary as soft. The dendritic cluster's low share is a different problem – 27 cells is too few to say much either way, and CellTypist splits them between a generic and a DC2 label.

The adjusted Rand index between the two labellings is 0.504, and that number should not be read as disagreement. CellTypist used 40 distinct labels on these cells against my 7, so the two partitions differ mostly in granularity: it splits my CD4 T cells into central-memory, effector-memory and naive subsets that my marker panel cannot separate. The per-cluster table is the meaningful comparison; the ARI mostly measures the resolution gap.

5 Limitations

- **One donor, one sample.** Nothing here speaks to between-individual variation or batch effects, which dominate real studies.
- **Annotation is manual**, though no longer unchecked: a classifier trained on a labelled atlas now agrees with every cluster label (Section 4.6). The rules themselves were still written after seeing the data, and CellTypist's own training data is not ground truth either.
- **Several choices were fixed, not tuned**: 15 PCs, 15 neighbours, 2,000 variable genes. I did not test sensitivity to them beyond clustering resolution and subsampling.
- **Rare populations** (megakaryocytes) are lost at the chosen resolution, and the stricter adaptive QC may remove some genuine low-count cells.
- **Reference labels are not ground truth.** The tutorial's labels came from a similar workflow on largely the same data.

6 Provenance and reproducibility

The data are public. All code is in `src/analysis.py`; every number and figure in this report comes from running it (`results.json`). The analysis and report were written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding the results. No part derives from university coursework or assessed material.

References

Domínguez Conde, C., Xu, C., Jarvis, L. B., Rainbow, D. B., Wells, S. B., Gomes, T., Howlett, S. K., Suchanek, O., Polanski, K., King, H. W., Mamanova, L., Huang, N., Szabo, P. A., Richardson, L., Bolt, L., Fasouli, E. S., Mahbubani, K. T., Prete, M., Tuck, L., … Teichmann, S. A. (2022). Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*, *376*(6594), eabl5197. https://doi.org/10.1126/science.abl5197

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